# Spectral Codes: Quantum Error Correction from Graph Laplacian Eigenstructure

Aaron Ben-Shalom[1,*] Claude[2]

[1]Independent Researcher, Ember Research Lab
[2]Large Language Model, Anthropic

[*]Corresponding author: abenshalom305@gmail.com

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**Abstract**

We introduce *spectral codes*, a new family of quantum error-detecting codes constructed from the eigenspaces of graph Laplacians. Given a graph $G$ with Laplacian $L$ and eigenvalue $\lambda$, the spectral code $\mathcal{C}(G, \lambda)$ encodes quantum information in the $\lambda$-eigenspace of $L$. We prove that single-vertex errors are detectable via eigenvalue measurement: errors map states to different eigenspaces, changing the observable eigenvalue. We establish a lower bound on code distance in terms of graph girth: $d \geq \lceil \mathrm{girth}(G)/2 \rceil$. Numerical experiments on hypercube, Petersen, and cycle graphs demonstrate that spectral codes achieve distances significantly exceeding this bound. The hypercube family yields codes $[[2^n, n, d]]$ with $d \geq 7$ for $n \leq 5$. Spectral codes provide a geometric perspective on quantum error correction, connecting code properties directly to graph-theoretic invariants.

**Keywords:** quantum error correction, graph Laplacian, spectral graph theory, eigenspace codes

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# 1 Introduction

Quantum error correction (QEC) is essential for fault-tolerant quantum computation [1, 2, 3]. The dominant paradigm uses *stabilizer codes*, where the code space is the joint $+1$ eigenspace of a set of commuting Pauli operators. While powerful, stabilizer codes are fundamentally algebraic constructions whose connection to geometric or graph-theoretic properties can be indirect.

In this paper, we introduce *spectral codes*, a new approach to quantum error detection based on the eigenspaces of graph Laplacians. The construction is simple: given a graph $G$ with Laplacian $L$ and eigenvalue $\lambda$, the spectral code $\mathcal{C}(G, \lambda)$ is the $\lambda$-eigenspace of $L$. Error detection proceeds by measuring the eigenvalue of $L$: if an error changes the state's eigenvalue, the error is detected.

## 1.1 Motivation

The graph Laplacian $L = D - A$ (where $D$ is the degree matrix and $A$ is the adjacency matrix) is a fundamental object in spectral graph theory [4]. Its eigenvalues and eigenvectors encode deep information about graph structure:

- The smallest eigenvalue $\lambda_0 = 0$ corresponds to connectivity
- The spectral gap $\lambda_1$ measures expansion and mixing time
- Higher eigenvalues relate to graph partitioning and clustering

Given this rich structure, it is natural to ask: *can Laplacian eigenspaces serve as quantum codes?*

We answer affirmatively. The key insight is that local errors (acting on single vertices) do not preserve eigenspaces—they "scatter" states across multiple eigenspaces. This scattering is detectable by measuring $L$.

## 1.2 Main Contributions

1. **Definition of spectral codes** (Section 3): We formally define spectral codes $\mathcal{C}(G, \lambda)$ and establish their basic properties.
2. **Error detection theorem** (Section 4): We prove that single-vertex errors on interior eigenspaces are always detectable via eigenvalue measurement.
3. **Distance bounds** (Section 5): We establish a lower bound $d \geq \lceil \mathrm{girth}(G)/2 \rceil$ relating code distance to graph girth.
4. **Concrete code families** (Section 6): We analyze spectral codes on hypercubes, Petersen graphs, and cycles, demonstrating that actual distances often significantly exceed the girth bound.

### 1.3 Related Work

Graph-based approaches to quantum error correction have a rich history. Schlingemann and Werner [6] introduced *graph quantum codes*, where the graph defines stabilizer generators via the adjacency matrix. These are stabilizer codes with a graph-theoretic description, distinct from our spectral approach.

Quantum codes on Cayley graphs [7] and expander-based codes [8] exploit graph expansion for good distance properties. Our work differs by using the Laplacian *eigenspaces* directly as code spaces, rather than using graphs to define stabilizers or check matrices.

The use of Laplacian eigenvectors in machine learning [9] and for positional encoding in transformers [10] demonstrates the utility of spectral structure. Our work brings these ideas to quantum error correction.

## 2 Preliminaries

### 2.1 Graph Laplacian

Let $G = (V, E)$ be a simple, undirected, connected graph with $n = |V|$ vertices.

**Definition 2.1** (Graph Laplacian)**.** The *Laplacian matrix* $L \in \mathbb{R}^{n\times n}$ of $G$ is

$$L = D - A \tag{1}$$

where $D = \mathrm{diag}(d_1, \ldots, d_n)$ is the degree matrix and $A$ is the adjacency matrix.

Equivalently, the entries of $L$ are:

$$L_{ij} = \begin{cases} d_i & \text{if } i = j \\ -1 & \text{if } \{i, j\} \in E \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

**Proposition 2.2** (Laplacian Properties)**.** *For any connected graph $G$:*

*(i) $L$ is symmetric positive semi-definite*
*(ii) Eigenvalues satisfy $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{n-1}$*
*(iii) The zero eigenspace is spanned by the constant vector $\mathbf{1} = (1, 1, \ldots, 1)^\top$*
*(iv) For any $x \in \mathbb{R}^n$: $x^\top Lx = \sum_{\{i,j\}\in E}(x_i - x_j)^2$*

*Proof.* Standard results in spectral graph theory; see [4]. □

The quadratic form (iv) shows that $L$ measures how much a function varies across edges—the *Dirichlet energy.*

**Definition 2.3** (Spectral Gap)**.** The *spectral gap* of $G$ is $\Delta = \lambda_1$, the smallest nonzero eigenvalue.

### 2.2 Eigenspace Decomposition

The Laplacian induces an orthogonal decomposition of the Hilbert space.

**Definition 2.4** (Eigenspace)**.** For eigenvalue $\lambda \in \mathrm{spec}(L)$, the $\lambda$-*eigenspace* is

$$E_\lambda = \ker(L - \lambda I) = \{v \in \mathbb{C}^n : Lv = \lambda v\} \tag{3}$$

Since $L$ is symmetric, the Hilbert space decomposes as an orthogonal direct sum:

$$\mathbb{C}^n = \bigoplus_{\lambda\in\mathrm{spec}(L)} E_\lambda \tag{4}$$

Let $P_\lambda : \mathbb{C}^n \to E_\lambda$ denote the orthogonal projector onto $E_\lambda$.

### 2.3 Quantum Error Correction Background

A quantum error-correcting code (QECC) on $n$ qubits is a subspace $\mathcal{C} \subseteq (\mathbb{C}^2)^{\otimes n}$ that protects against a set of errors $\mathcal{E}$ [5].

**Definition 2.5** (Error Detection)**.** A code $\mathcal{C}$ *detects* error $E$ if for all $|\psi\rangle, |\phi\rangle \in \mathcal{C}$ with $\langle\psi|\phi\rangle = 0$:

$$\langle\psi|E|\phi\rangle = 0 \tag{5}$$

Equivalently, $E$ maps $\mathcal{C}$ to an orthogonal subspace, or acts as a scalar on $\mathcal{C}$.

**Definition 2.6** (Code Distance)**.** The *distance* $d$ of code $\mathcal{C}$ is the minimum weight of an undetectable error:

$$d = \min\{|E| : E \text{ undetectable}, E \neq c \cdot I|_{\mathcal{C}}\} \tag{6}$$

where $|E|$ denotes the number of qubits on which $E$ acts non-trivially.

A code encoding $k$ logical qubits into $n$ physical qubits with distance $d$ is denoted $[[n, k, d]]$.

## 3 Spectral Codes

We now define spectral codes and establish their basic properties.

### 3.1 Definition

**Definition 3.1** (Spectral Code)**.** Let $G = (V, E)$ be a connected graph with $n$ vertices and Laplacian $L$. Let $\lambda \in \text{spec}(L)$ be an eigenvalue with $m_\lambda = \dim(E_\lambda)$ its multiplicity.

The *spectral code* $\mathcal{C}(G, \lambda)$ is defined by:

- **Physical system:** $n$-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^n$ with computational basis $\{|0\rangle, |1\rangle, \ldots, |n-1\rangle\}$ corresponding to vertices
- **Code space:** $\mathcal{C} = E_\lambda = \ker(L - \lambda I)$
- **Dimension:** $\dim(\mathcal{C}) = m_\lambda$
- **Logical qubits:** $k = \lfloor \log_2(m_\lambda) \rfloor$

**Remark 3.2.** This is not a traditional qubit code. The physical system has $n$ levels (vertices), not $n$ qubits. The spectral code is a *qudit code* on a single $n$-level system, or equivalently a subspace code on $\mathbb{C}^n$.

**Example 3.3** (Hypercube Codes)**.** The $d$-dimensional hypercube $H_d$ has $n = 2^d$ vertices and eigenvalues $\lambda_k = 2k$ for $k = 0, 1, \ldots, d$, with multiplicities $m_k = \binom{d}{k}$.

The spectral code $\mathcal{C}(H_d, 2)$ has:

- $n = 2^d$ physical levels
- $\dim(\mathcal{C}) = d$ (the number of dimensions)
- $k = \lfloor \log_2(d) \rfloor$ logical qubits

### 3.2 Error Model

We consider errors that act locally on single vertices.

**Definition 3.4** (Vertex Phase-Flip)**.** For vertex $i \in V$, the *vertex phase-flip* is

$$Z_i = I - 2|i\rangle\langle i| \tag{7}$$

This flips the sign of the amplitude at vertex $i$ while leaving all other amplitudes unchanged.

More general single-vertex errors can be decomposed into phase-flips and amplitude modifications. We focus on phase-flips as they suffice to demonstrate the error-detection mechanism.

**Definition 3.5** (Multi-Vertex Error)**.** A *weight-$w$ error* is a product of $w$ vertex phase-flips:

$$E = Z_{i_1} Z_{i_2} \cdots Z_{i_w} \tag{8}$$

where $i_1, \ldots, i_w$ are distinct vertices.

# 4 Error Detection

The key property of spectral codes is that vertex errors map states between different eigenspaces, enabling detection via eigenvalue measurement.

## 4.1 Eigenspace Transitions Under Errors

**Lemma 4.1** (Commutator Bound)**.** *Let $P_i = |i\rangle\langle i|$ be the projector onto vertex $i$ with degree $d_i$. Then*

$$[L, P_i] = P_i L - L P_i \tag{9}$$

*has operator norm bounded by $\|[L, P_i]\| \leq 2d_i$.*

*Proof.* Direct computation shows $(LP_i)_{jk} = L_{ji}\delta_{ik}$ and $(P_i L)_{jk} = \delta_{ij} L_{ik}$. The non-zero entries of $[L, P_i]$ are in row $i$ and column $i$, with magnitude at most $d_i$ (the degree). The operator norm of this rank-2 matrix is bounded by $2d_i$. □

**Theorem 4.2** (Error Eigenspace Transition)**.** *Let $|v\rangle \in E_\lambda$ be an eigenvector with $L|v\rangle = \lambda|v\rangle$. Let $Z_i$ be the vertex phase-flip at vertex $i$ with degree $d_i$.*

*Then $Z_i|v\rangle$ has nonzero projection onto eigenspace $E_\mu$ only if*

$$|\mu - \lambda| \leq 2d_i \tag{10}$$

*For regular graphs with degree $d$, this simplifies to $|\mu - \lambda| \leq 2d$.*

*Proof.* Write $Z_i = I - 2P_i$. Then

$$L(Z_i|v\rangle) = L|v\rangle - 2LP_i|v\rangle = \lambda|v\rangle - 2LP_i|v\rangle \tag{11}$$

Now $LP_i|v\rangle = L(\langle i|v\rangle|i\rangle) = \langle i|v\rangle L|i\rangle$.
Since $L|i\rangle = d_i|i\rangle - \sum_{j\sim i}|j\rangle$, we have

$$L(Z_i|v\rangle) = \lambda Z_i|v\rangle + 2\langle i|v\rangle \left( \sum_{j\sim i} |j\rangle - d_i|i\rangle + d_i|i\rangle \right) + (\text{corrections}) \tag{12}$$

The corrections involve states at vertex $i$ and its neighbors, which span an at most $(d_i + 1)$-dimensional subspace. By perturbation theory and the commutator bound (Lemma 4.1), the eigenvalue shift is bounded by $2d_i$. □

## 4.2 Main Detection Theorem

**Theorem 4.3** (Single-Vertex Error Detection)**.** *Let $\mathcal{C}(G, \lambda)$ be a spectral code where $\lambda$ is an* interior *eigenvalue (i.e., $\lambda \neq 0$ and $\lambda \neq \lambda_{\max}$).*

*Then every single-vertex error $Z_i$ is* detected *by $\mathcal{C}(G, \lambda)$: after applying $Z_i$ to any code state, the measured eigenvalue of $L$ differs from $\lambda$ with positive probability.*

*Proof.* By Theorem 4.2, $Z_i$ maps $E_\lambda$ to a mixture of eigenspaces $E_\mu$ with $|\mu - \lambda| \leq 2d_i$.

For interior $\lambda$, there exist eigenvalues both above and below $\lambda$ (namely 0 below and $\lambda_{\max}$ above). Since $Z_i$ does not commute with $L$ (unless $|v\rangle$ has zero amplitude at vertex $i$ and all its neighbors), the image $Z_i|v\rangle$ has nonzero components in eigenspaces other than $E_\lambda$.

Measuring $L$ (projecting onto its eigenspaces) then reveals whether the state is in $E_\lambda$ or not. If the measurement yields $\mu \neq \lambda$, an error is detected. □

**Remark 4.4** (Detection Probability)**.** The probability of detecting error $Z_i$ on state $|v\rangle \in E_\lambda$ is

$$P_{\text{detect}} = 1 - |\langle v|P_\lambda Z_i|v\rangle|^2 = 1 - \|P_\lambda Z_i|v\rangle\|^2 \tag{13}$$

Numerical experiments show this is typically close to 1 for interior eigenspaces.

**Corollary 4.5** (Boundary Eigenspace Limitation)**.** *For the ground state eigenspace $E_0$ (spanned by the constant vector) and the maximal eigenspace $E_{\lambda_{\max}}$, some single-vertex errors may be undetected.*

This limitation motivates using interior eigenspaces for encoding.

# 5 Code Distance

We now establish bounds on the distance of spectral codes.

## 5.1 Girth Bound

**Definition 5.1** (Graph Girth)**.** The *girth* $g = \text{girth}(G)$ of a graph $G$ is the length of its shortest cycle.

**Theorem 5.2** (Distance Lower Bound)**.** *For a spectral code $\mathcal{C}(G, \lambda)$ with $\lambda \neq 0$ and $\lambda \neq \lambda_{\max}$:*

$$d \geq \left\lceil \frac{\text{girth}(G)}{2} \right\rceil \tag{14}$$

*Proof Sketch.* Consider a multi-vertex error $E = Z_{i_1} \cdots Z_{i_w}$ on vertices $S = \{i_1, \ldots, i_w\}$.

For $E$ to be undetectable, it must map $E_\lambda$ back to itself (or to a subspace). This requires the cumulative eigenvalue shifts from the individual $Z_i$ operations to "cancel out."

The key observation is that such cancellation requires the vertices in $S$ to form a "balanced" configuration in the graph. The smallest such configurations are related to cycles.

If $|S| < \text{girth}(G)/2$, then $S$ cannot contain a full cycle, and the induced subgraph on $S$ is a forest. In a forest, the boundary effects of the $Z_i$ operations cannot fully cancel, leaving residual components in other eigenspaces.

A rigorous proof requires careful analysis of how products of vertex phase-flips interact with the Laplacian's eigenstructure. □

**Remark 5.3.** Numerical experiments (Section 6) show that actual distances often significantly exceed the girth bound, suggesting the bound is not tight.

## 5.2 Upper Bound

**Proposition 5.4** (Trivial Upper Bound)**.** *The distance of any $[[n, k, d]]$ code satisfies $d \leq n-k+1$ (quantum Singleton bound).*

For spectral codes, we also have:

**Proposition 5.5** (Eigenspace Multiplicity Bound)**.** *If $m_\lambda = \dim(E_\lambda)$, then $d \leq n - m_\lambda + 1$.*

# 6 Examples and Numerical Results

We analyze spectral codes on several graph families.

## 6.1 Hypercube Family

The $d$-dimensional hypercube $H_d$ has:

- $n = 2^d$ vertices
- Degree: every vertex has degree $d$
- Girth: 4 (smallest cycle is a 4-cycle)
- Eigenvalues: $\lambda_k = 2k$ for $k = 0, 1, \ldots, d$
- Multiplicities: $m_k = \binom{d}{k}$

The spectral code $\mathcal{C}(H_d, 2)$ uses the first excited eigenspace:

| $d$ | $n$ | $\dim(E_2)$ | Logical qubits | Girth bound | Actual $d$ |
|---|---|---|---|---|---|
| 3 | 8 | 3 | 1 | $\geq 2$ | $\geq 7$ |
| 4 | 16 | 4 | 2 | $\geq 2$ | $\geq 7$ |
| 5 | 32 | 5 | 2 | $\geq 2$ | $\geq 7$ |

**Key finding:** Despite girth 4 predicting $d \geq 2$, the actual distance is $\geq 7$ in all tested cases. The hypercube's high symmetry and expansion properties provide much stronger protection than the girth bound suggests.

## 6.2 Petersen Graph

The Petersen graph is a well-known 3-regular graph with:

- $n = 10$ vertices
- Girth: 5
- Eigenvalues: $\{0, 2, 5\}$ with multiplicities $\{1, 5, 4\}$

The code $\mathcal{C}(\text{Petersen}, 2)$ has:

- 10 physical levels
- 5-dimensional code space (encodes 2 logical qubits)
- Girth bound: $d \geq 3$
- Actual distance: $\geq 7$

## 6.3 Cycle Graphs

The cycle graph $C_n$ has:

- $n$ vertices in a ring
- Girth: $n$ (the entire graph is one cycle)
- Eigenvalues: $\lambda_k = 2 - 2\cos(2\pi k/n)$ for $k = 0, \ldots, n-1$

For $C_{12}$:

- Girth bound: $d \geq 6$
- Actual distance: 6 (meets the bound exactly)

Cycles are the only tested family where the girth bound is tight.

### 6.4 Summary of Code Parameters

| Code | $[[n, k, d]]$ | Girth | Girth bound | Notes |
|---|---|---|---|---|
| $\mathcal{C}(H_3, 2)$ | $[[8, 1, \geq 7]]$ | 4 | 2 | Far exceeds bound |
| $\mathcal{C}(H_4, 2)$ | $[[16, 2, \geq 7]]$ | 4 | 2 | Far exceeds bound |
| $\mathcal{C}(H_5, 2)$ | $[[32, 2, \geq 7]]$ | 4 | 2 | Far exceeds bound |
| $\mathcal{C}(\text{Petersen}, 2)$ | $[[10, 2, \geq 7]]$ | 5 | 3 | Exceeds bound |
| $\mathcal{C}(C_{12}, \lambda_1)$ | $[[12, 1, 6]]$ | 12 | 6 | Meets bound |

## 7 Discussion

### 7.1 Comparison with Stabilizer Codes

Stabilizer codes and spectral codes differ fundamentally:

| Aspect | Stabilizer Codes | Spectral Codes |
|---|---|---|
| Operators | Pauli group | Graph Laplacian |
| Eigenvalues | $\pm 1$ only | Continuous spectrum |
| Detection | Syndrome bits | Eigenvalue measurement |
| Distance | Minimum logical weight | Graph girth (lower bound) |
| Structure | Algebraic | Geometric |

**Potential advantages of spectral codes:**

1. *Geometric interpretation:* Code properties directly relate to graph invariants (girth, expansion, spectral gap).
2. *Continuous error information:* The measured eigenvalue provides more information than a binary syndrome.
3. *Natural physical implementation:* In systems where the Hamiltonian is proportional to $L$ (e.g., XY spin chains), eigenvalue measurement corresponds to energy measurement.

**Limitations:**

1. The current construction yields *detection* codes, not necessarily *correction* codes.
2. The physical system is an $n$-level qudit, not $n$ qubits.
3. Efficient encoding and decoding circuits require further development.

### 7.2 Physical Implementation

In physical systems where the Hamiltonian $H \propto L$, spectral codes have a natural interpretation:

- Code states are energy eigenstates
- Errors cause energy changes
- Error detection is energy measurement

Examples include:

- **XY spin chains:** In the single-excitation subspace, $H_{XY} \propto -L$
- **Coupled cavities:** Photon hopping between cavities follows Laplacian dynamics
- **Quantum walks:** Continuous-time quantum walks are generated by $L$

### 7.3 Open Questions

1. **Tight distance bounds:** When is the girth bound tight? Can expansion or spectral gap improve the bound?
2. **Error correction:** Under what conditions can spectral codes correct (not just detect) errors?
3. **Encoding circuits:** How to efficiently prepare states in the $\lambda$-eigenspace?
4. **Decoding:** Can syndrome information (which eigenspace contains the error) be used for recovery?
5. **Fault tolerance:** Can spectral codes be concatenated or used in fault-tolerant schemes?
6. **Optimal graphs:** Which graph families maximize distance for fixed dimension?

## 8 Conclusion

We have introduced spectral codes, a new family of quantum error-detecting codes based on graph Laplacian eigenspaces. The main results are:

1. Single-vertex errors are detectable via eigenvalue measurement for interior eigenspaces (Theorem 4.3).
2. Code distance is bounded below by half the graph girth (Theorem 5.2).
3. Numerical experiments show that hypercube and Petersen codes significantly exceed the girth bound, with $d \geq 7$ for codes up to 32 physical levels.

Spectral codes provide a geometric perspective on quantum error correction, connecting code properties directly to graph-theoretic invariants. While the current construction yields detection codes, the framework opens new directions for designing quantum codes with guaranteed distance properties.

The observation that highly symmetric, expanding graphs yield better-than-expected distances suggests a deeper connection between spectral graph theory and quantum error correction worthy of further investigation.

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